

EDUCATION

- **Cornell University** Ithaca, NY
PhD in Computer Science January 2021 - Present
- **Indian Institute of Technology (BHU) Varanasi** Varanasi, India
B. Tech. in Computer Science and Engineering - CGPA: 9.75/10.00 - Department Rank 1 July 2014 - May 2018

RESEARCH EXPERIENCE

- **Cornell University** Ithaca, NY
PhD Student January 2021 - June 2021
 - Advised by Dr. Rachit Agarwal.
 - **Understanding Host Network Stack Overheads** With CPU speeds evolving at a glacial pace compared to network interconnect bandwidth there is a concerted effort by the community to optimise or even replace the existing Linux network stack to fully utilise the potential of modern hardware. In this project, through extensive and thorough measurements, we analysed the Linux TCP stack to uncover the bottlenecks that currently exist and the impact of various software and hardware offloads. Using our insights we propose various recommendations relevant to the design of networking systems of the future. I designed and ran several experiments to measure the impact of NIC Rx descriptor size and TCP Rx buffer size on the end-to-end throughput and it's relation to the receiver cache miss rate in the presence of DDIO and it's relation to data-copy latency. This formed the basis of an important contribution in the paper. In addition, I was the primary contributor in preparing the final artifact of the publication.
- **Microsoft Research India** Bangalore, India
Research Fellow July 2018 - December 2020
 - Advised by Dr. Muthian Sivathanu and Dr. Ramchandran Ramjee.
 - **Gandiva_{fair}** is a set of Proportionally Fair Scheduling policies for Gandiva based on the concept of using job migration to balance the ticket-adjusted load across servers and then performing intraserver proportional share to achieve cluster-level proportional share. It transparently handles GPU heterogeneity using profiling and a novel GPU trading strategy to improve cluster efficiency while preserving fairness. I was involved in the design of the scheduling algorithm and policies and their implementation.
 - **Gandiva** is a Cluster Scheduler/Resource Manager for Deep Learning Training jobs utilizing CPU Scheduling primitives like time-slicing, migration, and profiling to schedule GPUs efficiently. I redesigned and implemented the entire cluster manager using **Kubernetes** to manage job containers and wrote the scheduling framework in **Scala** using the **Akka Actors** library for concurrency and **gRPC** for performing RPCs.
 - **GPU Proxy** is a mechanism inspired from **CRCUDA** to enable suspend/resume, and migration of Deep Learning jobs written in **PyTorch** and **TensorFlow**. I implemented **Horovod** and **NCCL** support to enable suspend/resume and migration of distributed jobs.

PUBLICATIONS

- Q. CAI, S. CHAUDHARY, M. VUPPALAPATI, J. HWANG, and R. AGARWAL. *Understanding Host Network Stack Overheads*. 2021. In Proceedings of the ACM Special Interest Group on Data Communication (SIGCOMM '21). <https://dl.acm.org/doi/abs/10.1145/3452296.3472888>
- S. CHAUDHARY, R. RAMJEE, M. SIVATHANU, N. KWATRA, and S. VISWANATHA. *Balancing Efficiency and Fairness in Heterogeneous GPU Clusters for Deep Learning*. 2020. In Proceedings of the Fifteenth European Conference on Computer Systems (EuroSys '20). <https://dl.acm.org/doi/abs/10.1145/3342195.3387555>

INTERNSHIP

- **Goldman Sachs**

Summer Analyst

Bangalore, India

May 2017 - July 2017

- **DbBench** is a distributed Database Benchmarking system for **SecDB**, a proprietary object database, used to make critical decisions about the future of the database system. It can emulate the real-life workload of hundreds of clients accessing a single database endpoint and report aggregate metrics like throughput and transactions per second. I designed and implemented the entire system in **Scala** using **Akka Actors** library for concurrency and **Netty** for composable high-throughput, low-latency asynchronous socket IO. I also implemented a **YAML** based configuration system and an interpreted Turing Complete programming language to write performance tests in.

SELECTED PROJECTS

- **Profiling Linux Network Stack** <https://github.com/Terabit-Ethernet/terabit-network-stack-profiling>
This repository contains the artifacts used to perform the experiments in our SIGCOMM '21 publication.
- **LTDP for Viterbi Algorithm** <https://github.com/5hubh4m/ltdp-viterbi-algorithm> Applied the Linear Tropical Dynamic Programming model to the Viterbi algorithm to parallelise the otherwise linear algorithm using the theorem of Rank Convergence. Implemented it using **OpenMPI** in conjunction with **OpenMP** in **C++**. Investigated the improvement over vanilla LTDP by redistributing data to processors after each principal iteration to prevent otherwise redundant computation of converged rows.
- **Scheme Interpreter in Haskell** <https://github.com/5hubh4m/simple-scheme> Wrote an Interpreter for a subset of Lisp (the R5RS standard) in **Haskell**, a pure functional programming language. Implemented an REPL, a parser, variables, user-defined functions using Functional Programming concepts like maps, folds, composition, monads, do-notation, currying, pattern-matching, etc.

TEACHING

- **Introduction to Computer Networks**

TA

Cornell University

February 2021 - May 2021

- Responsibilities included providing support and resolving queries during office hours, maintaining the course website, and grading assignments for a class of about 250 undergraduates.

- **Introduction to Computer Programming**

TA and Tutor

IIT (BHU) Varanasi

July 2016 - May 2018

- Responsibilities included delivering instruction during practical sessions on programming fundamentals in **C**, providing support and resolving queries during tutorial sessions, and curating and grading assignments for a class of about 60 undergraduates.

- **Artificial Intelligence**

TA

IIT (BHU) Varanasi

Jan 2018 - May 2018

- Responsibilities included delivering instruction during practical sessions on fundamentals of Artificial Intelligence (Search, Logic, and Machine Learning), curating assignments, and building and managing a **git** based encrypted automatic submission system for their evaluation for a class of about 80 undergraduates.

ACHIEVEMENTS

- **IIT (BHU) Varanasi Medal** for being Department Rank 1 in Computer Science and Engineering B. Tech Class of 2014-18.
- **SRFP 2016** Awarded the Summer Research Fellowship by the Indian Academy of Sciences, Bangalore.
- **Runner Up** at Microsoft's Code.Fun.Do 2016 24-Hrs Hackathon.
- **Winner** of Parliamentary Debate Competition at AMU Literary Festival, AMU Aligarh, 2015.

POSITIONS OF RESPONSIBILITY

- **Event Coordinator and Problem Curator** for Capture the Flag event at Technex 2017, and CodeFest 2017, with the participation of over 500 students from India and abroad.
- **Jt. Secretary** of the Literary Club of IIT (BHU) Varanasi for the session 2015-16.
- **Web Development** Worked on the Website Development and Technical Committee of Kashiyaatra 2016, FMC Weekend 2015, and IITBHUMUN 2015.